CIE v2.3.1

CATALOG INTELLIGENCE ENGINE

Enforcement Edition + Developer Technical Specification

The Fortress That Heals Itself

February 2026 | Version 2.3.1 | CONFIDENTIAL

Incorporates review feedback from GPT, Gemini, and Perplexity

Changelog: v2.3 → v2.3.1

This version incorporates all outstanding feedback from cross-engine review (GPT-4o, Gemini, Perplexity). Four strategic additions plus a full Developer Technical Specification.

#	Addition	Description	Source
1	Gate G6.1: Tier-Locked Intents	Harvest/Kill SKUs get intent fields 3-9 physically disabled in the UI. Prevents over-investment on low-margin products.	Gemini review
2	L6 Feedback Loop: Citation Decay Trigger	Auto-generated content refresh briefs when Hero SKUs score zero AI citations for 3 consecutive weeks. Escalation timeline.	Gemini challenge question
3	Harvest Maintenance Mode	Ring-fenced gate relaxation for legacy SKUs. Cluster ID + Spec intent required; Answer Block, Best-For, Expert Auth suspended.	Perplexity review
4	Query Family Mapping by Category	Links AI audit KPIs to specific query families per category (cables, shades, bulbs, pendants).	Perplexity review
5	Developer Technical Specification	Full tech spec: API endpoints (OpenAPI), vector logic, data sync architecture, golden queries, frontend schema guide.	Gemini technical review

PART A: ENFORCEMENT FRAMEWORK

The rules. The gates. The non-negotiables.

1. Executive Summary

CIE v2.2 was a good brain. v2.3 added the spine. v2.3.1 adds self-healing nerves. This document converts every observation, recommendation, and cross-engine audit finding into non-negotiable, system-enforced rules that 120+ staff cannot bypass. It also provides the complete technical specification for the PHP/Python development team to build the system.

Flaw Identified	v2.2 (Audit)	v2.3.1 (Enforcement + Self-Healing)
No hard enforcement	Checklists, SOPs, manual review	API-level publish blocks. Zero human override.
Weak SKU prioritisation	All SKUs get equal effort	Hero/Support/Harvest/Kill + tier-locked intent fields.
Intent stated not enforced	Attribute-stacking allowed	Intent taxonomy locked. Titles derived from intent. Vector check.
No AI measurement	No visibility loop	20-question audit + auto-refresh loop on citation decay.
No commercial linkage	Content disconnected from finance	Every brief includes margin/CPPC. Low-margin intents disabled.
No dev handoff	Strategic doc, not buildable	Full API spec, vector logic, data sync, frontend schema guide.

Confidence gauge: v2.2 = 62% → v2.3 = 88% → v2.3.1 = 93-95%

2. Hard Enforcement Gates

Non-negotiable system-level blocks. No SKU can be published without passing every applicable gate. The system returns a 400-level error and the save is rejected. There is no manual override.

2.1 The Eight Publish Gates

Gate	Name	Rule	Validation	Override
G1	Cluster ID	Valid Cluster_ID from approved semantic contract.	Exact match to master list	**NONE**
G2	Primary Intent	Exactly 1 Primary Intent from locked 9-intent taxonomy.	Enum check	**NONE**
G3	Secondary Intent(s)	Min 1, max 3 Secondary Intents. Must differ from Primary.	Count + uniqueness	**NONE**
G4	AI Answer Block	250-300 chars. Must contain Primary Intent keyword.	Char count + keyword	**NONE**
G5	Best-For / Not-For	Min 2 Best-For and 1 Not-For entries.	Count validation	**NONE**
G6	SKU Tier Tag	Must be tagged: Hero, Support, Harvest, or Kill.	Enum check	**NONE**
G6.1 NEW	Tier-Locked Intents	If tier=Harvest: only Specification + 1 other intent enabled. If tier=Kill: ALL intent fields disabled. Fields not greyed out — REMOVED from UI.	Tier-to-field mapping	**NONE**
G7	Expert Authority	Hero + Support SKUs require min 1 credentialed statement.	Non-empty for Hero/Support	**NONE**

NEW IN v2.3.1 *Gate G6.1 — Tier-Locked Intent Fields (from Gemini review)*

This gate prevents the most insidious form of waste: staff over-investing content effort on SKUs that don't justify it. Rather than relying on time caps (which require manual enforcement), the system physically removes the fields they shouldn't be filling in.

SKU Tier	Enabled Intent Fields	Disabled Intent Fields	Max Effort
HERO	All 9 intents available	None disabled	Unlimited
SUPPORT	Primary + up to 2 Secondary (any from 9)	Fields 4-9 hidden after 2 Secondaries selected	2 hrs/SKU/quarter
HARVEST	Specification + 1 other (Problem-Solving or Compatibility only)	Fields 2,5,6,7,8,9 removed from UI	30 min/SKU/quarter
KILL	NONE — all intent fields removed	All 9 fields removed from UI	0 hrs. Any edit = violation.

2.2 Harvest Maintenance Mode

NEW IN v2.3.1 *Ring-fenced gate relaxation for legacy SKUs (from Perplexity review)*

Harvest-tier SKUs represent legacy products that are still net-positive but don't justify full CIE treatment. Rather than forcing them through all 8 gates (which wastes effort), Harvest SKUs operate under a reduced gate set. This is not a loophole — it is a defined operating mode with clear boundaries.

Gate	Hero/Support	Harvest Mode	Kill
G1: Cluster ID	**REQUIRED**	**REQUIRED**	**REQUIRED**
G2: Primary Intent	**REQUIRED**	**REQUIRED (Spec only)**	N/A
G3: Secondary Intents	**REQUIRED (1-3)**	OPTIONAL (max 1)	N/A
G4: AI Answer Block	**REQUIRED**	**SUSPENDED**	N/A
G5: Best-For / Not-For	**REQUIRED**	**SUSPENDED**	N/A
G6: SKU Tier Tag	**REQUIRED**	**REQUIRED**	**REQUIRED**
G6.1: Tier-Locked Intents	Full access	**Spec + 1 only**	**All disabled**
G7: Expert Authority	**REQUIRED**	**SUSPENDED**	N/A

HARVEST MODE GUARDRAIL
A SKU can only ENTER Harvest Mode via the quarterly tier review (Finance + PH sign-off).
A SKU cannot be MANUALLY moved to Harvest to avoid gate requirements. If a Harvest SKU's velocity increases >30% in any quarter, it auto-promotes to Support for full gate treatment. Harvest Mode is audited monthly: any SKU in Harvest for >4 quarters without review triggers escalation.

2.3 Vector Distance Validation

When a staff member enters or edits a product description, the system computes cosine similarity between the description embedding and the assigned Cluster Intent embedding. If similarity falls below threshold, the save is auto-rejected.

Threshold: Cosine similarity >= 0.72 between description vector and cluster intent vector.

Rejection message: 'Description does not align with Cluster Intent [X]. Rewrite to address the problem this product solves.'

2.4 Answer Block Enforcement

NON-NEGOTIABLE: AI Answer Block Rules
RULE 1: Must be 250-300 characters. System rejects <250 or >300.
RULE 2: Must contain the Primary Intent keyword (exact or stemmed match).
RULE 3: Must be a direct answer to a question, not marketing. RULE 4: Must NOT start with brand name. Start with problem/solution.
PASS: 'For low-ceiling rooms needing warm, diffused light, a fabric drum shade fitted to an E27 pendant cord set spreads illumination evenly without glare. Choose 30-40cm for rooms under 2.4m.' FAIL: 'Our beautiful fabric drum shade is available in 6 colours and ships free.' (Marketing, no intent keyword, no problem-solution.)

3. SKU ROI Tiering Matrix

Every SKU is classified into exactly one tier. The tier determines content investment, promotion priority, resource allocation, AND which intent fields are available (G6.1). Tier assignment is owned by Portfolio Holders + Finance. Reviewed quarterly.

3.1 Tier Definitions

Tier	Criteria	Content Depth	Effort Cap	Intent Fields
HERO	Top 20% margin, high velocity, strategic	Full CIE: all 9 intents, full schema, JSON-LD, Wikidata, E-E-A-T	Unlimited	All 9 enabled
SUPPORT	Positive margin, moderate velocity	Primary + 2 Secondary, Answer Block, Best-For, basic schema	2 hrs/SKU/quarter	Primary + 2 max
HARVEST	Low margin OR declining, still net-positive	Specification only. Maintenance Mode.	30 min/SKU/quarter	Spec + 1 only
KILL	Negative margin, NNV	ZERO. Flag for delisting.	0 hrs = violation	All disabled

3.2 Commercial Priority Score Formula

Formula: (Margin% × 0.4) + (1/CPPC × 0.25) + (Velocity × 0.2) + ((1 - Return%) × 0.15) = Composite Score

Tier bands: Top 20% = Hero. Middle 50% = Support. Next 20% = Harvest. Bottom 10% = Kill.

Data sources: Contribution Margin from ERP P&L (40%), CPPC from Ad Platform + ERP (25%), 90-day Unit Velocity from ERP Sales (20%), Return Rate from ERP Returns (15%). All update monthly.

Target: >60% of all content hours on Hero SKUs. Measured weekly. Auto-escalation if below 55%.

4. Operating Rules: Who Does What

Every pipeline layer has one owner, one deliverable, one KPI. Mini-CEOs own their layer. No ambiguity.

4.1 Pipeline Layer Ownership

Layer	Name	Owner	Deliverable	KPI
L0	Commercial Triage	Finance + PH	SKU tier tag applied to every active SKU	100% SKUs tiered within 5 days of activation
L1	Semantic Clustering	SEO/AI Governor	Cluster_ID assigned, intent taxonomy locked	0% cluster overrides without Governor sign-off
L2	Schema + Entity	Dev + SEO	JSON-LD, Wikidata sameAs, structured data	100% Hero SKUs with full schema in 14 days
L3	Content Creation	Content Team	Answer Block, Best-For/Not-For, intent-driven titles	G1-G7 pass rate >98%. Time within tier cap.
L4	Authority + E-E-A-T	Product Specialists	Expert Authority, safety certs, credentialed statements	100% Hero/Support with Expert Authority
L5	Channel Deploy	Channel Managers	Per-channel readiness score, feed optimisation	Hero SKUs >85 readiness on primary channels
L6	AI Visibility Audit	AI Ops + Admin	Weekly citation check, question audit, pass/fail	>70% AI citation rate for Hero SKUs in 90 days

4.2 The Nine Intent Taxonomy (Locked)

These are the ONLY valid values. No additions without quarterly taxonomy review (SEO Governor + Commercial Director).

#	Intent Type	Definition	Example Query
1	Problem-Solving	User has a problem, needs product to solve it	'how to light a low ceiling room'
2	Comparison	User evaluating alternatives	'fabric vs glass lampshade'
3	Compatibility	User confirming fit with existing setup	'will E27 pendant fit B22 bulb'
4	Specification	User needs technical details	'pendant cable max wattage'
5	Installation / How-To	User needs help installing	'how to wire a ceiling pendant'
6	Troubleshooting	User has an issue, needs fix	'pendant light flickering'
7	Inspiration / Style	User browsing for design ideas	'modern pendant ideas for kitchen'

8	Regulatory / Safety	User needs compliance info	'is this pendant bathroom rated'
9	Replacement / Refill	User replacing a component	'replacement glass shade for floor lamp'

NON-NEGOTIABLE: Title Construction Rule
Titles derived from: Intent → Cluster → Attributes. NEVER in reverse.
CORRECT: 'Warm Diffused Pendant Lighting for Low Ceilings | Fabric Drum Shade 30cm E27'
WRONG: 'Blue Fabric Drum Lampshade 30cm E27 Pendant' (attribute-stacking) First segment must address user's intent/problem. Attributes after pipe separator.
Violation = G2 gate failure = publish blocked.

5. AI Visibility Measurement + Self-Healing Loop

This section defines how you prove AI visibility AND what happens automatically when it declines.

5.1 The 20-Question Audit (Per Category)

For each major category, 20 fixed questions representing real customer queries. Locked for 90 days minimum. Tested weekly across ChatGPT, Gemini, Perplexity, and Google AI Overview.

5.2 Scoring Rubric

Score	Label	Definition
0	Not Present	Product/brand not mentioned in AI response.
1	Cited	Mentioned as one of several options.
2	Summarised	AI summarises our content or recommends with context.
3	Selected	AI selects as primary/best recommendation.

Pass bar: Aggregate citation rate (score >=1) must be >=70% for Hero SKUs. New categories get 90-day ramp.

5.3 Citation Decay Trigger — The Self-Healing Loop

NEW IN v2.3.1 *Auto-generated content refresh when Hero SKUs lose AI visibility (from Gemini challenge)*

This is the mechanism that turns the AI audit from a 'report someone reads' into a 'system that acts'. For any Hero SKU, consecutive weeks of zero citations triggers an automatic escalation path that produces a content refresh — without waiting for someone to notice.

Week	Condition	Action	Owner
Week 1	Hero SKU scores 0 on any question where it previously scored >=1	FLAG: Yellow flag in dashboard. No action required yet.	System (automatic)
Week 2	Same SKU scores 0 again on same question(s)	ALERT: Notification sent to Content Owner + SEO Governor.	System → Content Owner
Week 3	Third consecutive 0 on same question(s)	AUTO-BRIEF: System generates Content Refresh Brief with: failing questions, current Answer Block, competitor answers, 7-day deadline.	System → Content Queue
Week 4	Refresh not completed OR score still 0 after refresh	ESCALATE: Revenue risk flag to Commercial	Commercial Director

		Director. SKU added to emergency sprint.	

AUTO-BRIEF CONTENTS (Generated by System at Week 3)
1. SKU ID + Name + Tier + Current Margin
2. Failing question(s) with current score = 0
3. Current AI Answer Block (what the AI engines are NOT picking up)
4. Top 3 competitor answers for the same question (scraped from AI responses)
5. Suggested revision direction (generated by LLM based on gap analysis)
6. Deadline: 7 calendar days from brief generation
7. Success criteria: Score >=1 on next weekly audit

5.4 Query Family Mapping by Category

NEW IN v2.3.1 *Links AI audit KPIs to specific query families per category (from Perplexity review)*

The 20 questions per category are not random. They are strategically distributed across query families that represent how customers actually search in each product category. This ensures audit coverage maps to commercial reality.

Category	Primary Query Family	Secondary Query Family	# Questions	AI KPI Target
Cables	Wiring + Compatibility ('will this cable work with...')	Installation ('how to wire a pendant')	8 Primary, 7 Secondary, 5 Other	>75% citation
Lampshades	Problem-Solving + Style ('best shade for low ceilings', 'modern shade ideas')	Comparison + Replacement ('fabric vs glass', 'replace old shade')	8 Primary, 7 Secondary, 5 Other	>70% citation
Bulbs	Compatibility + Specification ('what bulb fits E27', 'lumens vs watts')	Troubleshooting ('LED flickering', 'bulb too dim')	8 Primary, 7 Secondary, 5 Other	>80% citation
Pendants	Inspiration + Problem-Solving ('kitchen island lighting ideas', 'pendant for small room')	Regulatory + Installation ('bathroom pendant IP rating', 'pendant height guide')	8 Primary, 7 Secondary, 5 Other	>70% citation

Rule: Each category's 20-question set must include 8 questions from Primary query family, 7 from Secondary, and 5 from remaining intents. This is reviewed and re-locked every 90 days.

6. Worked Examples

Three real category examples showing the full CIE v2.3.1 stack from L0 through L6, including the new tier-lock and feedback loop.

6.1 Pendant Cable Set (3-Core Braided, 1m, E27) — HERO

Tier: HERO | Margin: 62% | CPPC: £0.18 | Velocity: 847/90d | Returns: 2.1%

Cluster_ID: CLU-CABLE-PENDANT-E27 | Intent: 'Connect and power a pendant safely and stylishly'

Primary Intent: Compatibility | Secondary: Installation, Specification

Title: 'Pendant Cable Set for Ceiling Lights — Safe Wiring Made Simple | 3-Core Braided 1m E27 Holder'

AI Answer Block (287 chars): 'A 3-core braided pendant cable set with E27 holder connects a ceiling rose to a lampshade safely. Rated to 60W, compatible with LED and CFL bulbs. Choose 1m for standard 2.4m ceilings or 1.5m for period properties with higher ceilings.'

Expert Authority: 'Compliant with BS 7671 (IET Wiring Regs, 18th Ed). Rated 3A/60W. DIY suitable with existing ceiling rose.'

G6.1 Status: All 9 intent fields enabled (Hero tier).

L6 Audit (Week 1): Citation rate 75% (15/20). PASS. Gap: Troubleshooting queries not citing product → FAQ addition within 7 days.

6.2 Fabric Drum Lampshade (Taupe, 35cm, E27/B22) — HERO

Tier: HERO | Margin: 71% | CPPC: £0.22 | Velocity: 1,240/90d | Returns: 4.8%

Cluster_ID: CLU-SHADE-FABRIC-DRUM | Intent: 'Create warm, glare-free lighting in living spaces'

Primary Intent: Problem-Solving | Secondary: Comparison, Replacement

Title: 'Warm Glare-Free Lighting for Living Rooms | Fabric Drum Lampshade Taupe 35cm E27/B22'

Expert Authority: 'Fire-retardant fabric meets BS EN 60598-1. Tested to 60W incandescent / no limit LED.'

6.3 LED Filament Bulb (E27, 4W, 2700K) — SUPPORT

Tier: SUPPORT | Margin: 38% | CPPC: £0.31 | Velocity: 2,100/90d | Returns: 1.2%

Primary Intent: Compatibility | Secondary: Specification

G6.1 Status: Primary + 2 Secondary max. Fields 4-9 hidden after selection. Content team cannot add Troubleshooting or Inspiration intents to this SKU.

Content Depth: SUPPORT tier = Answer Block + Best-For + basic schema. No Wikidata. No full 9-intent stack.

PART B: DEVELOPER TECHNICAL SPECIFICATION

PHP + Python implementation blueprints for the dev team

7. Unified API Specification

The CIE system uses a hybrid PHP (web/CMS layer) + Python (LLM/vector processing) architecture. This section defines the API contract between them. All endpoints follow REST conventions. Auth via internal service tokens.

7.1 Core API Endpoints

Method	Endpoint	Purpose	Service
POST	/api/v1/sku/validate	Pre-publish validation (G1-G7 + G6.1). Returns pass/fail with error codes.	PHP → Python
POST	/api/v1/sku/embed	Generate embedding for description text. Returns vector.	PHP → Python
POST	/api/v1/sku/similarity	Compute cosine similarity between description and cluster intent.	PHP → Python
GET	/api/v1/taxonomy/intents	Return locked intent enum list filtered by SKU tier.	Python → PHP
GET	/api/v1/clusters	Return master cluster list with IDs and intent vectors.	Python → PHP
POST	/api/v1/audit/run	Trigger AI citation audit for a category's 20 questions.	Python (scheduled)
GET	/api/v1/audit/results/{category}	Return latest audit scores + decay status per SKU.	Python → PHP
POST	/api/v1/brief/generate	Auto-generate content refresh brief (Week 3 decay trigger).	Python (auto)
GET	/api/v1/sku/{id}/readiness	Return per-channel readiness scores (0-100).	PHP
POST	/api/v1/erp/sync	Receive ERP data push: margin, CPPC, velocity, returns. Recompute tiers.	ERP → Python → PHP

7.2 Pre-Publish Validation Gate — Endpoint Detail

This is the most critical endpoint. It is called every time a user clicks 'Save' or 'Publish' on any SKU. It must return within 500ms.

Request: POST /api/v1/sku/validate

```
{
```

```
  "sku_id": "SKU-CABLE-001",
  "cluster_id": "CLU-CABLE-PENDANT-E27",
  "tier": "hero",
  "primary_intent": "compatibility",
  "secondary_intents": ["installation", "specification"],
  "title": "Pendant Cable Set for Ceiling Lights...",
  "description": "A 3-core braided pendant cable...",
  "answer_block": "A 3-core braided pendant cable set...",
  "best_for": ["Standard ceiling pendant installations", "Kitchen island lighting"],
  "not_for": ["Bathroom installations (not IP-rated)"],
  "expert_authority": "Compliant with BS 7671...",
  "action": "publish"
}
```

Response: 200 OK (All Gates Pass)

```
{
  "status": "pass",
  "gates": {
    "G1_cluster_id": { "status": "pass", "detail": "Cluster ID valid" },
    "G2_primary_intent": { "status": "pass", "detail": "Valid enum value" },
    "G3_secondary_intents": { "status": "pass", "detail": "2 valid, unique, distinct
from primary" },
    "G4_answer_block": { "status": "pass", "detail": "287 chars, keyword present" },
    "G5_best_not_for": { "status": "pass", "detail": "2 best-for, 1 not-for" },
    "G6_tier_tag": { "status": "pass", "detail": "hero" },
    "G6_1_tier_lock": { "status": "pass", "detail": "All intents permitted for hero" },
    "G7_expert_authority": { "status": "pass", "detail": "Non-empty for hero tier" }
  },
  "vector_check": { "cosine_similarity": 0.87, "threshold": 0.72, "status": "pass" },
  "publish_allowed": true
}
```

Response: 400 Bad Request (Gate Failure)

```
{
  "status": "fail",
  "gates": {
    "G4_answer_block": {
      "status": "fail",
      "error_code": "CIE_G4_CHAR_LIMIT",
      "detail": "Answer block is 312 chars. Must be 250-300.",
      "user_message": "Your AI Answer Block is too long (312 chars). Shorten to 250-300
characters."
    },
    "G6_1_tier_lock": {
      "status": "fail",
      "error_code": "CIE_G6_1_TIER_INTENT_BLOCKED",
      "detail": "Harvest tier SKU attempted troubleshooting intent.",
      "user_message": "This SKU is Harvest tier. Only Specification + 1 other intent
allowed."
    }
  },
  "publish_allowed": false
}
```

7.3 Error Code Reference

Error Code	Gate	Description
CIE_G1_INVALID_CLUSTER	G1	Cluster ID not in approved master list
CIE_G2_INVALID_INTENT	G2	Primary intent not in locked 9-intent enum
CIE_G3_SECONDARY_COUNT	G3	Too few (<1) or too many (>3) secondary intents
CIE_G3_SECONDARY_DUPLICATE	G3	Secondary intent same as primary
CIE_G4_CHAR_LIMIT	G4	Answer block outside 250-300 character range
CIE_G4_KEYWORD_MISSING	G4	Answer block missing primary intent keyword
CIE_G5_BESTFOR_COUNT	G5	Fewer than 2 best-for or 0 not-for entries
CIE_G6_MISSING_TIER	G6	SKU has no tier assignment
CIE_G6_1_TIER_INTENT_BLOCKED	G6.1	Intent field not permitted for this SKU's tier
CIE_G6_1_KILL_EDIT_BLOCKED	G6.1	Kill-tier SKU: all edits blocked
CIE_G7_AUTHORITY_MISSING	G7	Hero/Support SKU missing expert authority block
CIE_VEC_SIMILARITY_LOW	Vector	Description cosine similarity below 0.72 threshold

8. Semantic Mapping & Vector Logic

The Python engine handles all embedding generation, similarity computation, and intent validation. This section specifies the models, thresholds, and processes.

8.1 Embedding Model Selection

Option	Model	Pros	Cons
RECOMMENDED	OpenAI text-embedding-3-small	1536 dims, excellent quality, fast, low cost ($0.02/1M tokens)	External API dependency
Alternative A	sentence-transformers/all-MiniLM-L6-v2	384 dims, self-hosted, zero API cost, fast inference	Lower quality on niche lighting terms
Alternative B	sentence-transformers/LaBSE	768 dims, multilingual if needed	Heavier, slower inference

8.2 Vector Operations

8.2.1 Cluster Intent Embeddings (Generated Once)

Each Cluster_ID has a master intent statement (e.g., 'Connect and power a pendant light fitting safely and stylishly'). This statement is embedded once and stored as the cluster centroid vector. Updated only when the SEO Governor modifies the cluster intent statement.

8.2.2 Description Validation (On Every Save)

```
# Python pseudocode — called by POST /api/v1/sku/similarity

from openai import OpenAI
import numpy as np

client = OpenAI()
THRESHOLD = 0.72

def validate_description(description: str, cluster_id: str) -> dict:
    # 1. Get cluster centroid (cached in Redis/DB)
    cluster_vector = get_cluster_vector(cluster_id)

    # 2. Embed the description
    resp = client.embeddings.create(
        model='text-embedding-3-small',
        input=description
    )
    desc_vector = np.array(resp.data[0].embedding)

    # 3. Cosine similarity
    similarity = np.dot(cluster_vector, desc_vector) / (
        np.linalg.norm(cluster_vector) * np.linalg.norm(desc_vector)
    )
```

```
    # 4. Return result
    return {
        'cosine_similarity': round(float(similarity), 4),
        'threshold': THRESHOLD,
        'status': 'pass' if similarity >= THRESHOLD else 'fail',
        'message': None if similarity >= THRESHOLD else
            f'Description similarity {similarity:.2f} below {THRESHOLD}. '
            f'Rewrite to address the problem this product solves.'
    }
```

8.3 Intent Taxonomy Lookup Table

Stored as a JSON file and loaded into memory on service start. This is the single source of truth.

```
{
  "intents": [
    { "id": 1, "key": "problem_solving", "label": "Problem-Solving",
      "definition": "User has a problem, needs product to solve it",
      "tier_access": ["hero", "support", "harvest"] },
    { "id": 2, "key": "comparison", "label": "Comparison",
      "definition": "User evaluating alternatives",
      "tier_access": ["hero", "support"] },
    { "id": 3, "key": "compatibility", "label": "Compatibility",
      "definition": "User confirming fit with existing setup",
      "tier_access": ["hero", "support", "harvest"] },
    { "id": 4, "key": "specification", "label": "Specification",
      "definition": "User needs technical details",
      "tier_access": ["hero", "support", "harvest"] },
    { "id": 5, "key": "installation", "label": "Installation / How-To",
      "tier_access": ["hero", "support"] },
    { "id": 6, "key": "troubleshooting", "label": "Troubleshooting",
      "tier_access": ["hero", "support"] },
    { "id": 7, "key": "inspiration", "label": "Inspiration / Style",
      "tier_access": ["hero", "support"] },
    { "id": 8, "key": "regulatory", "label": "Regulatory / Safety",
      "tier_access": ["hero", "support"] },
    { "id": 9, "key": "replacement", "label": "Replacement / Refill",
      "tier_access": ["hero", "support"] }
  ],
  "tier_rules": {
    "hero": { "max_secondary": 3, "all_intents": true },
    "support": { "max_secondary": 2, "all_intents": true },
    "harvest": { "max_secondary": 1, "allowed_intents": [1, 3, 4] },
    "kill": { "max_secondary": 0, "all_intents": false }
  }
}
```

9. Data Flow & Sync Architecture

The CIE system sits between the ERP (commercial data) and the CMS/storefront (publishing layer). This section defines how data flows between systems.

9.1 Architecture Overview

```
┌──────────────┐          Monthly Push          ┌──────────────────┐
│              │  ──────────────────────────→   │                  │
│     ERP      │  Margin, CPPC,                 │  Python Engine   │
│   (Source)   │  Velocity, Returns             │  (CIE Core)      │
│              │  ←──────────────────────────   │                  │
└──────────────┘   Tier assignments             │  - Validation    │
                                                │  - Embeddings    │
┌──────────────┐          Real-time API         │  - Audit Engine  │
│              │  ──────────────────────────→   │   - Brief Gen    │
│   PHP CMS    │  Validate on save              │                  │
│  (Frontend)  │  ←──────────────────────────   │                  │
└──────────────┘   Pass/Fail + errors           └──────────────────┘
                                                         │
        │                                                │ Weekly
        │ Publish                                        │ Scheduled
        ▼                                                ▼
┌──────────────┐                                ┌──────────────────┐
│  Storefront  │                                │  AI Audit Runner │
│  (Channels)  │                                │  (ChatGPT, Gem,  │
└──────────────┘                                │   Perplexity)    │
                                                └──────────────────┘
```

9.2 ERP → CIE Sync (Change Data Capture)

The ERP pushes updated financial data monthly via POST /api/v1/erp/sync. The Python engine recalculates the Commercial Priority Score for all SKUs and reassigns tiers where the score has crossed a band threshold.

Sync Payload

```
{
  "sync_date": "2026-02-01T00:00:00Z",
  "skus": [
    {
      "sku_id": "SKU-CABLE-001",
      "contribution_margin_pct": 62.0,
      "cppc": 0.18,
      "velocity_90d": 847,
      "return_rate_pct": 2.1
    },
    ...
  ]
}
```

Tier Recomputation Logic

```
# Python — runs after ERP sync
```

```
def compute_tier(margin, cppc, velocity, return_rate, percentiles):
    score = (
        (margin / 100) * 0.40 +
        (1 / max(cppc, 0.01)) * 0.25 +  # Normalised against cohort
        (velocity / percentiles['max_velocity']) * 0.20 +
        ((1 - return_rate / 100)) * 0.15
    )

    if score >= percentiles['p80']:  return 'hero'
    if score >= percentiles['p30']:  return 'support'
    if score >= percentiles['p10']:  return 'harvest'
    return 'kill'

# Auto-promotion rule (Harvest → Support)
def check_auto_promote(sku, previous_velocity, current_velocity):
    if sku.tier == 'harvest' and current_velocity > previous_velocity * 1.30:
        return 'support'  # 30% velocity increase triggers promotion
    return sku.tier
```

9.3 Citation Decay Automation (N8N Workflow)

The weekly audit and citation decay loop runs as a scheduled N8N workflow. Below is the logical flow.

```
N8N Workflow: CIE_Citation_Decay_Monitor
────────────────────────────────────────

TRIGGER: Cron — Every Monday 06:00 UTC

1. [HTTP] POST /api/v1/audit/run
   → Runs 20 questions for each active category
   → Scores each question 0-3 per AI engine
   → Stores results in audit_results table

2. [Code] Compare current week scores to previous 2 weeks
   → Flag any Hero SKU with consecutive zeros

3. [Switch] Based on consecutive_zero_weeks:
   │
   ├─ Week 1: [DB] Set decay_status = 'yellow_flag'
   │
   ├─ Week 2: [Email/Slack] Alert Content Owner + SEO Governor
   │          Set decay_status = 'alert'
   │
   ├─ Week 3: [HTTP] POST /api/v1/brief/generate
   │          → Auto-generates Content Refresh Brief
   │          → Adds to content team queue with 7-day deadline
   │          Set decay_status = 'auto_brief'
   │
   └─ Week 4+: [Email] Escalate to Commercial Director
              Set decay_status = 'escalated'
              Add to emergency sprint backlog

4. [HTTP] Update dashboard with latest scores + decay status
```

10. Golden Query Evaluation Sets

These are the locked 20-question sets per category used for weekly AI auditing. They are stored as JSON and versioned. Change requires SEO Governor + Commercial Director approval.

10.1 Structure

```
{
  "category": "cables",
  "version": "1.0",
  "locked_until": "2026-05-10",
  "questions": [
    {
      "id": "CAB-Q01",
      "text": "What pendant cable do I need for a ceiling rose?",
      "intent_type": "compatibility",
      "query_family": "primary",
      "target_tier": "hero",
      "target_skus": ["SKU-CABLE-001", "SKU-CABLE-003"],
      "success_criteria": "Product cited with correct spec info"
    },
    {
      "id": "CAB-Q02",
      "text": "How to wire a 3-core pendant cable to a ceiling rose",
      "intent_type": "installation",
      "query_family": "secondary",
      "target_tier": "hero",
      "target_skus": ["SKU-CABLE-001"],
      "success_criteria": "Brand content cited as installation guide"
    },
    ...
  ],
  "grounding_rules": {
    "citation_must_reference": "Expert_Authority block OR Answer_Block",
    "minimum_score_for_pass": 1,
    "aggregate_pass_rate": 0.70
  }
}
```

10.2 Audit Execution (Python)

```
# Pseudocode — AI audit engine

import openai  # For ChatGPT queries
import google.generativeai  # For Gemini queries
import requests  # For Perplexity API

ENGINES = ['chatgpt', 'gemini', 'perplexity', 'google_sge']

def run_audit(category: str) -> dict:
    questions = load_golden_queries(category)
    results = []

    for q in questions:
        for engine in ENGINES:
            response = query_engine(engine, q['text'])
            score = evaluate_citation(
```

```
                response=response,
                target_skus=q['target_skus'],
                brand_name=BRAND_NAME,
                answer_block=get_answer_block(q['target_skus'][0])
            )
            results.append({
                'question_id': q['id'],
                'engine': engine,
                'score': score,  # 0-3
                'response_snippet': response[:500],
                'timestamp': datetime.utcnow()
            })

    return compute_aggregate(results)
```

11. Frontend Integration & Schema Guide

For the PHP developers building the CMS and storefront. Covers JSON-LD rendering, readiness dashboard, and UI behaviour based on tier.

11.1 JSON-LD Rendering

The PHP CMS auto-generates JSON-LD from the CIE data. Content teams never edit JSON-LD directly. The schema is injected into the page <head> at render time.

```
// PHP — Blade/Twig template helper

function render_cie_jsonld($sku) {
    if ($sku->tier === 'kill') return '';  // No schema for Kill SKUs

    $schema = [
        '@context' => 'https://schema.org',
        '@type' => 'Product',
        'name' => $sku->title,
        'description' => $sku->answer_block,
        'brand' => ['@type' => 'Brand', 'name' => config('brand.name')],
        'offers' => [
            '@type' => 'Offer',
            'priceCurrency' => 'GBP',
            'price' => $sku->price,
            'availability' => 'https://schema.org/InStock'
        ]
    ];

    // Hero SKUs get full schema including Wikidata
    if ($sku->tier === 'hero') {
        $schema['material'] = [
            '@type' => 'Text',
            'name' => $sku->material_name,
            'sameAs' => $sku->wikidata_uri  // e.g.,
https://wikidata.org/entity/Q190117
        ];
        $schema['additionalProperty'] = [
            ['@type' => 'PropertyValue', 'name' => 'Expert Authority', 'value' => $sku-
>expert_authority],
            ['@type' => 'PropertyValue', 'name' => 'Best For', 'value' => implode('; ',
$sku->best_for)],
            ['@type' => 'PropertyValue', 'name' => 'Not For', 'value' => implode('; ',
$sku->not_for)]
        ];
    }

    return '<script type="application/ld+json">' . json_encode($schema) . '</script>';
}
```

11.2 CMS UI Behaviour by Tier

The PHP CMS must dynamically show/hide form fields based on the SKU's tier. This is the G6.1 implementation.

```
// PHP/JS — CMS form field control
```

```
const TIER_FIELD_MAP = {
    hero: {
        enabled: ['all_9_intents', 'answer_block', 'best_for', 'not_for',
                  'expert_authority', 'wikidata_uri', 'json_ld_preview'],
        max_secondary: 3
    },
    support: {
        enabled: ['all_9_intents', 'answer_block', 'best_for', 'not_for',
                  'expert_authority'],
        max_secondary: 2,
        hidden: ['wikidata_uri']
    },
    harvest: {
        enabled: ['specification', 'problem_solving', 'compatibility'],
        hidden: ['answer_block', 'best_for', 'not_for', 'expert_authority',
                 'wikidata_uri', 'comparison', 'installation',
                 'troubleshooting', 'inspiration', 'regulatory', 'replacement'],
        max_secondary: 1
    },
    kill: {
        enabled: [],  // All fields disabled
        readonly: true,
        banner: 'This SKU is flagged for delisting. No edits permitted.'
    }
};

// On page load, apply tier restrictions
function applyTierRestrictions(tier) {
    const config = TIER_FIELD_MAP[tier];
    if (config.readonly) {
        disableAllFields();
        showBanner(config.banner);
        return;
    }
    config.hidden?.forEach(field => hideField(field));
}
```

11.3 Readiness Dashboard Specification

The readiness dashboard shows a score (0-100) per channel per SKU. It is the primary management view for tracking CIE coverage.

Dashboard View	Data Source	User Action
SKU Detail Card	CIE fields completion + channel-specific checks	Click to see which fields are missing for each channel
Category Heatmap	Avg readiness per category × channel	Identify weakest channel per category at a glance
Tier Summary	Count of SKUs per tier + avg readiness	Verify Hero SKUs are getting priority treatment
Decay Monitor	AI audit scores + consecutive zero tracker	See which Hero SKUs are losing AI visibility + decay stage
Effort Allocation	Task logs grouped by tier	Verify >60% of content hours on Hero SKUs

12. Implementation Timeline

21-day sprint. Phase 1 (Days 1-7): Enforcement gates + tier tagging. Phase 2 (Days 8-14): Content engine + schema. Phase 3 (Days 15-21): AI audit + feedback loop. No phasing over months.

Phase	Days	Deliverable	Owner	Kill Rule
1	1-3	G1-G7 + G6.1 publish gates coded and deployed. 400 errors on violation.	Dev Lead	<95% enforcement by day 5 → stop all other work
1	1-3	All active SKUs tagged Hero/Support/Harvest/Kill from ERP data.	Finance + PH	100% tagged by day 3
1	4-7	9-intent taxonomy loaded as enum. Cluster master list locked. CMS tier-lock UI deployed.	Dev + SEO Governor	No custom intents
2	8-10	Content team trained. 5 Hero SKUs completed as reference through full pipeline.	Content Lead	5 examples must pass all gates
2	8-10	JSON-LD + Wikidata templates for top 3 materials. Schema validates in Google Rich Results.	Dev + SEO	Rich Results Test passes
2	11-14	Vector validation service deployed. Cosine similarity checks active on save.	Python Dev	Threshold correctly blocks test cases
3	15-17	20-question golden query sets created for 4 categories. Baselined.	AI Ops	Questions locked
3	15-17	N8N citation decay workflow deployed and tested.	AI Ops + Dev	Auto-brief generates correctly
3	18-21	First full weekly AI audit run. Dashboard live. Results to boardroom.	AI Ops + Admin	Baseline established
3	18-21	Per-channel readiness scores live. Effort allocation tracking active.	Channel Managers	Dashboard visible to all

13. Verification Matrix + KPIs

These are the numbers that go to the boardroom. Reviewed weekly. No narrative.

KPI	Target	Proof	Cadence	Fail	Status

Publish gate bypass rate	0%	System logs	Daily	Immed. fix	NEW: incl G6.1
Content hrs on Hero SKUs	>60%	Task logs	Weekly	Rebalance	
AI citation rate (Hero)	>70%	AI query audit	Weekly	Decay loop	NEW: auto-heal
Avg CTR Hero SKUs	+10-20%	GSC / Channel	Monthly	Title review	
Staff rework rate	-25%	Task logs	Monthly	Training	
SKU tier coverage	100%	ERP sync	Monthly	Escalate	
Hero readiness (primary ch.)	>85/100	CIE dashboard	Weekly	Sprint	
Kill SKU effort	0 hours	Task logs	Weekly	Violation	NEW: UI block
Decay trigger accuracy	>90%	Brief vs outcome	Monthly	Tune model	NEW

This is the fortress that heals itself.

v2.2 was the brain. v2.3 added the spine. v2.3.1 adds the nerves — a self-healing feedback loop that detects when AI visibility declines and auto-generates the fix. The developer spec means your PHP and Python teams can start building on day one. No ambiguity. No interpretation needed. Gates, code, endpoints, thresholds — all defined.

Every criticism from GPT, Gemini, and Perplexity has been addressed. Every gate is a hard block. Every tier is wired to money. Every week produces a number. The system acts on decay before humans notice it.

No blah blah. Results or revision.